Attention Heads are Complementary Visual Units: Mitigating Hallucinations in LVLMs via Adaptive Visual Cues Focusing

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Abstract

Recent studies have explored attention dynamics in Large Vision-Language Models (LVLMs). However, most existing approaches aggregate attention across heads, potentially obscuring head-specific visual information and weakening visual grounding, leading to hallucinations. In this work, we revisit hallucination from the perspective of how visual information is distributed and utilized across attention heads. We observe that a small subset of visual tokens accounts for most of the attention within each attention head, with these tokens—defined as head-wise visual cues—being complementary across heads, suggesting that effective grounding requires preserving head-specific support. We further find that the proportion of visual cues within visual attention declines during generation, leading to key visual information loss and hallucination. In addition, hallucinated tokens show weaker utilization of long-range textual context compared to correctly generated tokens. Building on these findings, we propose **V**isual Cu**e**s **R**einforcement via Head-Adaptive **D**e-redundancy and Distance-Aware **A**ttenuation **(VerDA)**, a training-free method to mitigate hallucinations. Specifically, we remove redundant visual tokens via head-adaptive de-redundancy, suppress local textual bias through distance-aware attenuation, and reinforce visual cues by redistributing attention. Experiments across multiple LVLMs and benchmarks demonstrate that VerDA consistently outperforms existing methods in mitigating hallucinations, with negligible inference overhead. The code is available at https://github.com/huazhenglin2003/VerDA.

1 Introduction

Large Vision-Language Models (LVLMs) [1, 2, 3, 4, 5] achieve significant progress in various vision-language tasks such as image captioning [6] and visual question answering [7]. However, LVLMs still face the challenge of hallucination [8, 9, 10], where the generated text is inconsistent with the visual input. This problem significantly limits their deployment in real-world applications [11, 12].

To mitigate hallucinations, several approaches incorporate external knowledge bases for alignment training [13, 14] or fine-tuning [15]. However, these methods often incur high computational and storage costs. In contrast, recent studies have focused on training-free approaches during the inference

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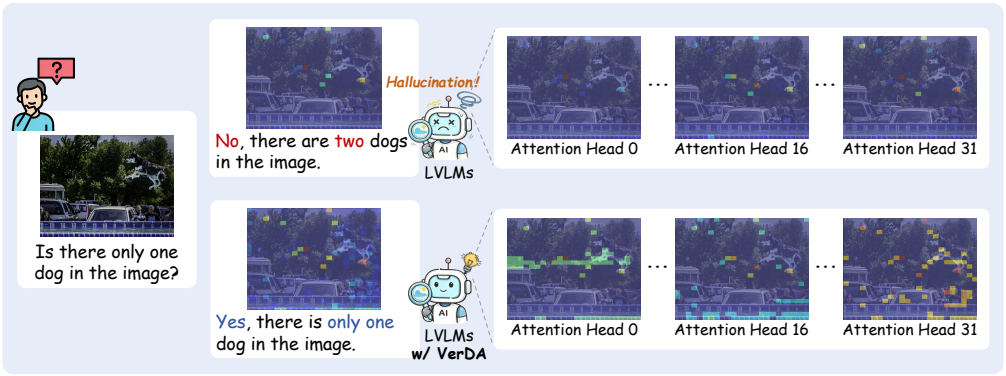


Figure 1: VerDA mitigates hallucinations by improving head-adaptive visual cues utilization.

stage. One line of research focuses on decoding strategies [16, 17, 18, 19, 20], which adjust the logits distribution during generation to suppress language priors. However, these methods mainly operate at the logits level rather than the internal visual attention mechanism, failing to effectively promote visual grounding and limiting their effectiveness in mitigating hallucinations. Another line of research analyzes hallucination from the perspective of attention patterns [21, 22, 23, 24, 25, 26], which enhance or redistribute visual attention during decoding. Although some of these methods have studied visual attention in the head, the sparsity of visual cues within a single head and their complementarity between heads have not been fully explored, which limits accurate visual grounding and their ability to mitigate hallucinations. Furthermore, they often characterize visual forgetting coarsely as a decrease in visual attention as generation proceeds [27, 28, 29], failing to capture the change of key visual information.

To address these issues, we analyze hallucination in LVLMs from the perspective of differences across attention heads and visual cues. Specifically, we observe that a small subset of visual tokens accounts for most of the attention within each attention head. We define these high attention visual tokens as the visual cues of each head, whose functional relevance is supported by the retention results in Figure 2 (d), where head-wise selection consistently outperforms global selection, as well as by the complementary masking analysis in Appendix A, where removing these selected tokens causes the largest performance degradation. Meanwhile, we further observe that different attention heads capture complementary visual cues, forming a structured pattern of visual information distribution (see ***Finding 1***). We further investigate the phenomenon of visual forgetting from the perspective of visual cues. Our analysis suggests that visual forgetting is not only reflected by a decrease in total visual attention, but also by a decline in the proportion of visual cues within visual attention, weakening the model's effective use of visual cues during generation (see ***Finding 2***). Finally, from a textual perspective, we introduce the Distance-Weighted Attention Mean (DWAM), a metric that provides a diagnostic signal for insufficient long-range textual context utilization. Through DWAM, we observe that hallucination in LVLMs is often associated with a distance-decay effect, where the model shows limited ability to utilize distant contextual information (see ***Finding 3***).

Motivated by these findings, we propose a training-free method to mitigate hallucinations, **V**isual Cue**s** Reinforcement via Head-Adaptive **D**e-redundancy and Distance-Aware **A**ttenuation (VerDA). Specifically, we first design a head-adaptive visual de-redundancy to remove low-value visual tokens within each head. Then, we introduce a distance-aware text attenuation mechanism based on DWAM to suppress over-reliance on local textual context. Finally, we propose a cues reallocation strategy that redistributes redundant attention to high-value visual cues in a head-adaptive manner, aiming to mitigate visual forgetting. Figure 1 shows how VerDA mitigates hallucinations by improving head-adaptive visual cues utilization. Extensive experiments across multiple LVLMs and benchmarks show that our method reduces hallucination while improving visual understanding. Our main contributions are summarized as follows:

- We identify three findings related to hallucination in LVLMs: (i) visual cues are sparse within each head, while different heads capture complementary visual cues; (ii) visual forgetting can be further characterized by a decrease in the proportion of visual cues; (iii) we introduce the DWAM metric that reflects the model's ability to utilize long-range context, and observe that weaker utilization of such context is associated with hallucination.

- We propose VerDA, a training-free method that mitigates hallucinations while alleviating visual forgetting by head-adaptive de-redundancy, while jointly suppressing over-reliance on local textual context and reinforcing key visual cues.
- Extensive experiments demonstrate that VerDA outperforms existing methods on widely-adopted hallucination benchmarks with negligible additional time cost.

2 Our Insights

Finding 1: Visual cues are sparsely distributed within each attention head, while different heads capture complementary visual cues. To characterize this phenomenon, we introduce head discriminability, which measures the difference of primary visual regions between a head and others. Specifically, given the visual attention distribution $a_{t,i}^{(l,h)}$ is the attention weight at the h-th head in the l-th layer to visual token i at step t, we first define the corresponding minimal coverage set $\mathcal{I}_{l,h}$ as:

$$\mathcal{I}_{l,h} = \arg\min_{\mathcal{I}_{l,h} \subseteq \mathcal{V}} |\mathcal{I}_{l,h}|, \quad \text{s.t.} \quad \sum_{i \in \mathcal{I}_{l,h}} a_{t,i}^{(l,h)} \geq \rho \sum_{i \in \mathcal{V}} a_{t,i}^{(l,h)}, \tag{1}$$

where $\mathcal{V}$ denotes the set of visual tokens, and ρ denotes the attention coverage ratio. Based on this minimal coverage set, the discriminability of head h is defined as:

$$D_h = \frac{1}{L} \sum_{l=0}^{L-1} \left(1 - \frac{1}{H-1} \sum_{\substack{h' \in [0,\cdots,H-1] \\ h' \neq h}} \frac{|\mathcal{I}_{l,h} \cap \mathcal{I}_{l,h'}|}{|\mathcal{I}_{l,h} \cup \mathcal{I}_{l,h'}|} \right), \tag{2}$$

where L and H denote the number of large language model layers and attention heads, respectively. A higher D_h indicates more distinct visual regions attended by the h-th head. We further define the union coverage across heads as:

$$R_{\text{union}} = \frac{1}{L} \sum_{l=0}^{L-1} \frac{\left| \bigcup_{h=0}^{H-1} \mathcal{I}_{l,h} \right|}{|\mathcal{V}|}. \tag{3}$$

We conduct an experiment on a random sample of 1,000 images from the CHAIR benchmark [8]. As shown in Figure 2 (b), we report the average visual token ratio $|\mathcal{I}_{l,h}|/|\mathcal{V}|$ per head under the coverage constraint ($\rho = 0.6$) and the corresponding union coverage across heads. Each head selects only 7.8% of visual tokens on average, while their union covers 44.9% of the visual space. This indicates that informative visual regions are sparse within each head while exhibiting complementarity across heads. Detailed per-head results are provided in Figure 2 (c), which show consistent trends.

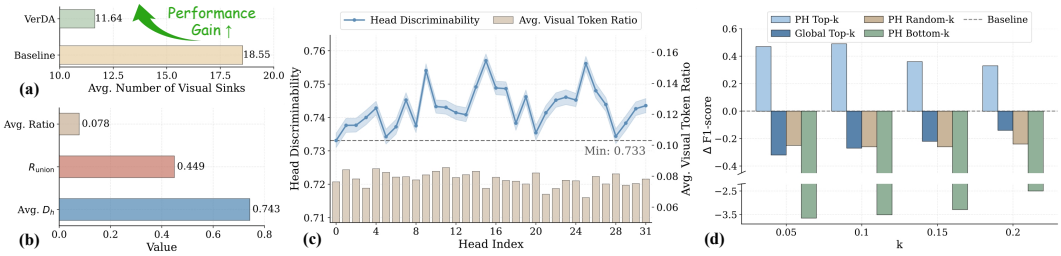


Figure 2: **Left (a):** Comparison of the average number of layer-level visual attention sinks between the baseline and VerDA. **Left (b):** Summary statistics of head-wise visual cues, including the average visual token ratio, union coverage ratio R_{union}, and average head discriminability D_h. **Middle (c):** Head-wise discriminability and the average token ratio required to cover 60% of attention within each head. **Right (d):** Interventional analysis of visual cues by retaining different subsets of visual tokens on the POPE benchmark across different retention ratios k. The dashed line marks the baseline ($\Delta\text{F1} = 0$), and lower ΔF1 indicates more severe hallucination. PH denotes per-head.

To further examine whether the head-wise high attention visual tokens play a functional role in hallucination mitigation, we compare different visual token retention strategies on the POPE benchmark [9].

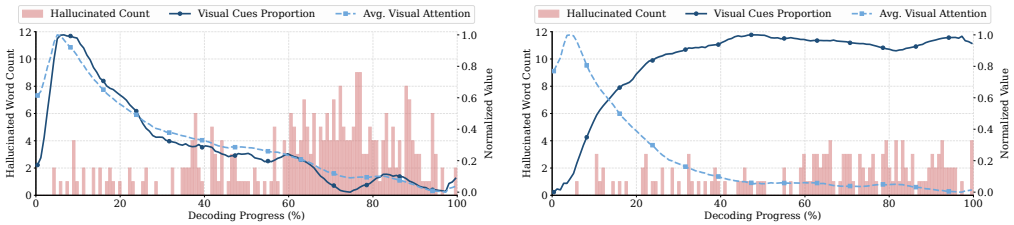


Figure 3: Analysis of visual forgetting during decoding. **Left (a):** As decoding progresses, both total visual attention and the proportion of visual cues decrease, accompanied by an increase in hallucination. **Right (b):** Under a controlled setting where total visual attention is kept constant, increasing the attention proportion toward visual cues consistently reduces hallucination.

Specifically, the per-head Top-k strategy retains the highest attention visual tokens independently for each attention head, while per-head Random-k and Bottom-k retain randomly selected and the lowest attention visual tokens, respectively. As shown in Figure 2 (d), per-head Top-k consistently outperforms Global Top-k across different retention ratios, indicating that globally aggregated attention fails to capture the head-specific visual information required for reliable grounding. Moreover, per-head Top-k clearly surpasses Random-k and Bottom-k, and even improves over the baseline when retaining only 10% of visual tokens, suggesting substantial redundancy in visual tokens and the effectiveness of head-wise selection. Appendix A further confirms the necessity of selected head-wise Top-k tokens via a complementary masking analysis. Finally, we further analyze layer-level visual attention sinks [22]. Figure 2 (a) shows that applying VerDA consistently reduces the number of layer-level visual sinks. This suggests that the complementary high-value visual information retained across heads alleviates the excessive concentration of redundant attention within layers.

Finding 2: The proportion of visual cues within visual attention decreases during generation, which can further characterize visual forgetting. As decoding progresses, the number of text tokens increases while the number of visual tokens remains fixed, naturally leading to a modality imbalance and a relative decline in overall visual attention. This suggests that the decrease in total visual attention may be partially attributed to textual sequence growth. Therefore, characterizing visual forgetting as a decrease in visual attention as generation proceeds is insufficient to fully capture the phenomenon. Building on Finding 1, we further investigate visual forgetting on the same CHAIR benchmark samples. Specifically, we examine the relationship among total visual attention, the proportion of visual cues within visual attention, and the hallucination throughout decoding. As shown in Figure 3(a), both total visual attention and the proportion of visual cues consistently decrease with decoding steps. Furthermore, as shown in Figure 3(b), when controlling similar trends in total visual attention through redistributing visual attention to concentrate on visual cues, we observe that increasing the relative proportion of visual cues corresponds to reduced hallucination. This suggests that visual forgetting can be further characterized by the decreasing proportion of visual cues.

Finding 3: Hallucination in LVLMs is associated with a distance-decay effect in textual context utilization. Beyond visual degradation, we further observe a distance-decay effect in textual context utilization. Although prior work [30, 31, 32] has noted that LVLMs often exhibit a bias toward local textual context during generation, such observations have remained at the level of describing context utilization patterns, and their link to hallucination remains unclear. To this end, we introduce Distance-Weighted Attention Mean (DWAM), a simple and interpretable metric to provide a diagnostic signal for insufficient long-range textual context utilization. The DWAM at decoding step t is defined as:

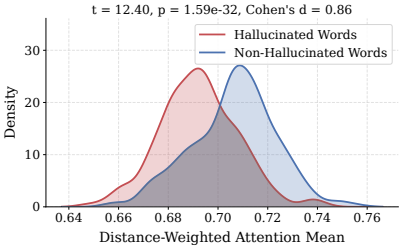


Figure 4: Relationship between DWAM and hallucinated word generation.

$$\text{DWAM}_t = \frac{1}{L}\frac{1}{H}\sum_{l=0}^{L-1}\sum_{h=0}^{H-1}\sum_{j\in X_{\mathcal{T}}} a_{t,j}^{l,h}\cdot d_{t,j}, \quad d_{t,j} = \frac{|j-(|X_{\mathcal{T}}|-1)|}{\max(|X_{\mathcal{T}}|-1,\epsilon)}, \tag{4}$$

where $X_{\mathcal{T}}$ denotes the visible textual context, $\epsilon > 0$ is a small constant, $a_{t,j}^{(l,h)}$ is the attention weight at the h-th head in the l-th layer to text token j at step t and $d_{t,j}$ is normalized distance between

token j and the current decoding position. A larger DWAM indicates that the model allocates more attention to distant textual context. Using the same CHAIR samples as in Finding 1, we analyze the relationship between DWAM and hallucination. As shown in Figure 4, words associated with hallucinations exhibit lower DWAM than non-hallucinated words, indicating weaker utilization of long-range contextual information. This difference is statistically significant, as confirmed by a t-test and Cohen's d ($t = 12.40$, $p = 1.59 \times 10^{-32}$, $d = 0.86$).

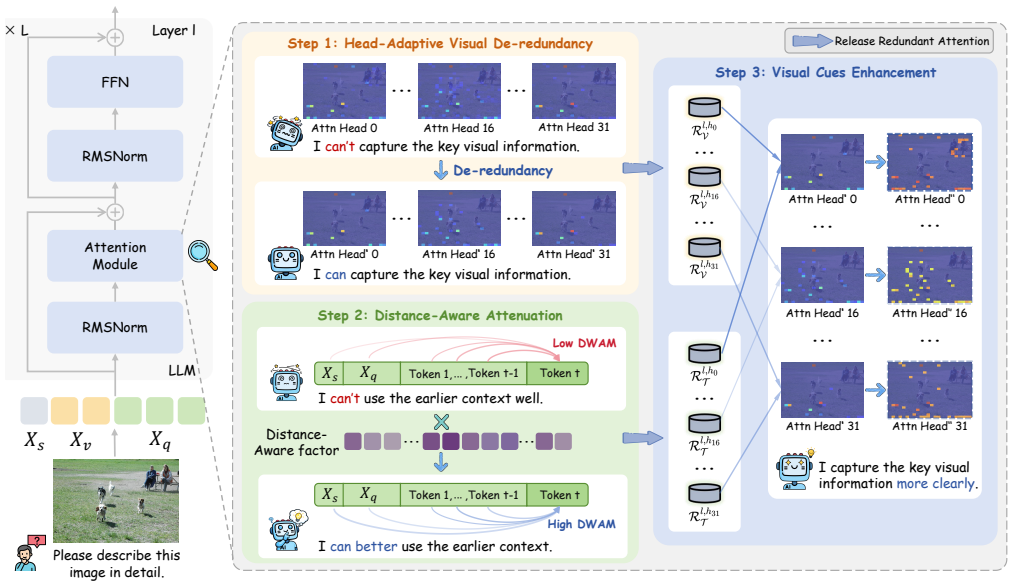


Figure 5: Overview of the proposed method VerDA. We remove redundant visual tokens in a head-adaptive manner, attenuate textual bias based on the DWAM metric, and reallocate attention to key visual cues to improve grounded generation for a given layer.

3 Methodology

3.1 Preliminaries

LVLM generation. LVLMs typically consist of a vision encoder, a modality connector, and an autoregressive language model. Given a system prompt, an input image, and a textual query, the image is first encoded into vision tokens X_v by the vision encoder and then projected to the text embedding space through a connector. Meanwhile, the system prompt and textual query are tokenized into text tokens, denoted as X_s and X_q, respectively. These tokens are then combined and passed into the LLM for autoregressive generation:

$$y_t = \arg\max p_\theta(y_t \mid y_{<t}, X_s, X_v, X_q), \tag{5}$$

where y_t and $y_{<t}$ denote the currently and the previously generated text tokens, respectively.

Multi-head attention. Multi-head attention (MHA) mechanism is a fundamental component of Transformer architectures. Let $X^{(l)} \in \mathbb{R}^{N \times d}$ denote the input to the l-th layer. For the h-th attention head in the l-th layer, the attention weights are computed as:

$$A^{(l,h)}(X^{(l)}) = \text{softmax}\left(\frac{Q^{(l,h)}(K^{(l,h)})^\top}{\sqrt{d_k}} + M\right), \tag{6}$$

where $Q^{(l,h)} = X^{(l)}W_Q^{(l,h)}$ and $K^{(l,h)} = X^{(l)}W_K^{(l,h)}$ denote the query and key matrices obtained from the learned projection matrices, respectively. M is the causal mask and d_k is the dimension of the query (Q) and key (K) vectors. The MHA mechanism concatenates the outputs of all the attention heads and projects them to form the layer output:

$$\text{MHA}^{(l)}(X^{(l)}) = \text{Concat}_{h=0}^{H-1}\left(A^{(l,h)}V^{(l,h)}\right)W_O^{(l)}, \tag{7}$$

where $V^{(l,h)} = X^{(l)} W_V^{(l,h)}$ denotes the value matrix obtained from the learned projection matrix. $W_O^{(l)}$ is the learned weight matrix for the output linear transformation.

3.2 Head-Adaptive Visual De-redundancy

Motivated by the observation that low-attention regions often contain redundant information (Finding 1), we formulate visual cues selection as a minimal subset coverage problem. Specifically, we select the minimal subset of visual tokens that covers a target ratio of the total visual attention:

$$\begin{aligned} \mathcal{I}_{l,h}^* &= \arg\min_{\mathcal{I}_{l,h} \subseteq \mathcal{V}} |\mathcal{I}_{l,h}|, \\ \text{s.t.} \quad \sum_{i \in \mathcal{I}_{l,h}} a_{t,i}^{(l,h)} &\geq \rho \sum_{i \in \mathcal{V}} a_{t,i}^{(l,h)}, \quad |\mathcal{I}_{l,h}| \geq \lceil \gamma \cdot |\mathcal{V}| \rceil, \quad \forall h \in \{1, \ldots, H\}, \end{aligned} \tag{8}$$

where $\rho \in (0, 1]$ is the coverage ratio, and $\gamma \in (0, 1]$ is the minimum retention ratio to prevent over-aggressive removal. Based on this head-adaptive selection, the de-redundant visual attention for the h-th head in the l-th layer is constructed as:

$$\hat{a}_{t,i}^{(l,h)} = \begin{cases} a_{t,i}^{(l,h)}, & i \in \mathcal{I}_{l,h}^*, \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

The released visual attention is given by: $R_{t,\mathcal{V}}^{(l,h)} = \sum_{i \in \mathcal{V} \setminus \mathcal{I}_{l,h}^*} a_{t,i}^{(l,h)}$. This design ensures that each attention head retains the minimal number of tokens necessary to preserve sufficient visual cues, thereby eliminating redundancy while avoiding excessive loss of informative signals.

3.3 Distance-Aware Attenuation

Beyond visual redundancy, our analysis in Finding 3 reveals a distance-decay effect on the text side. To mitigate this issue, we introduce a distance-aware attenuation mechanism over textual attention. Specifically, we first concatenate the prefix text tokens $X_{\mathcal{P}}$ and suffix text tokens $X_{\mathcal{S}}$ in their original order to form a unified text sequence $X_{\mathcal{T}} = [X_{\mathcal{P}}, X_{\mathcal{S}}]$, as visual tokens are typically interleaved within the text sequence in multimodal inputs. Inspired by DWAM, we further define a cross-head aggregated distance-saliency factor at the step t for a text token at position j in the l-th layer:

$$\bar{s}_j^{(l)} = \sigma\left(\sum_{h=0}^{H-1} a_{t,j}^{(l,h)}\right) \cdot \left(\lambda + (1-\lambda) \cdot d_j^{(l)}\right), \quad d_j^{(l)} = \frac{|j - (|X_{\mathcal{T}}| - 1)|}{\max(|X_{\mathcal{T}}| - 1, \epsilon)}, \quad j \in \mathcal{T}, \tag{10}$$

where $\sigma(\cdot)$ denotes the sigmoid function, $\mathcal{T}$ denotes the set of text tokens, and $\lambda \in (0, 1)$ controls the minimum retention ratio. We set $\lambda = 0.5$ to prevent over-attenuation of distant tokens. However, this shared mechanism does not account for the varying degrees of visual grounding across different heads. To address this, we introduce a head-specific gating factor based on the remaining visual cues $g_h^{(l)}$. Combining the shared distance–saliency factor and the head-specific gate, the attenuated textual attention is given by:

$$\hat{a}_{t,j}^{(l,h)} = a_{t,j}^{(l,h)} \cdot \bar{s}_j^{(l)} \cdot g_h^{(l)}, \quad \text{where } g_h^{(l)} = 1 - \sum_{i \in \mathcal{I}_{l,h}^*} a_{t,i}^{(l,h)}. \tag{11}$$

The released textual attention is given by: $R_{t,\mathcal{T}}^{l,h} = \sum_{j \in \mathcal{T}} \left(a_{t,j}^{(l,h)} - \hat{a}_{t,j}^{(l,h)}\right)$. This design jointly considers token-level relevance and head-specific visual grounding, enabling effective suppression of local textual bias while preserving informative context.

3.4 Visual Cues Head-Adaptive Enhancement

After visual de-redundancy and text attenuation, we further reallocate the released attention to reinforce significant visual cues, enhancing the contribution of grounded visual content during generation and mitigating visual forgetting (Finding 2). For each attention head in the l-th layer, the refined attention is then defined as:

$$\tilde{a}_{t,i}^{(l,h)} = \begin{cases} \hat{a}_{t,i}^{(l,h)} + \frac{R_{t,\mathcal{V}}^{(l,h)} + R_{t,\mathcal{T}}^{(l,h)}}{|\mathcal{I}_{l,h}^*|}, & i \in \mathcal{I}_{l,h}^*, \\ \hat{a}_{t,i}^{(l,h)}, & \text{otherwise.} \end{cases} \tag{12}$$

This design enables effective reuse of the released attention, leading to enhanced visual cues without increasing the overall attention budget. The complete procedure of VerDA is provided in Algorithm 1 in Appendix B. An overview of the proposed framework is illustrated in Figure 5.

4 Experiments

Models and Baselines. To demonstrate the broad applicability of VerDA across LVLM architectures and scales, we evaluate it on a diverse set of recent models, including LLaVA-1.5-7B [2], LLaVA-NeXT-8B [33], InstructBLIP-13B [3], Qwen2.5-VL-7B and Qwen3-VL-8B [4, 34], InternVL2-8B, InternVL3-9B/14B [35, 36, 5], and GLM-4V-9B [37]. We compare against a range of decoding and attention-based baselines, including Greedy decoding, Beam search, CODE [19], VCD [16], OPERA [18], DoLa [17], DropoutDecoding [20], VAF [21], SPARC [24], VHR [23], and AdaIAT [25]. More details are provided in Appendix C.

Benchmarks. We evaluate VerDA on seven benchmarks with different focuses: hallucination benchmarks (CHAIR [8], POPE [9], AMBER [38] and MMHal-Bench [39]), a comprehensive benchmark (MME [40]), and generation quality benchmarks (CapMAS [41], LLaVA-Bench [2]). More details are provided in Appendix D.

Implementation Details. We set $\gamma = 0.1$ for all experiments and $\rho = 0.6$ for most models, except InstructBLIP where $\rho = 0.1$ due to its Q-Former-based visual compression. VerDA is applied to a fixed subset of layers for each model series and used consistently across all benchmarks, with detailed configurations provided in Appendix E.1. We faithfully reproduced all baseline methods based on their open-source repositories and set the hyperparameters according to the values reported in the papers. The results of all methods are reported under consistent conditions of base models, prompts, and generation parameters to ensure a fair comparison. Specifically, the maximum number of generated tokens is set to 512, and the beam size is set to 5.

4.1 Evaluation Results on Hallucination Benchmarks.

CHAIR and POPE Evaluations. Table 1 and Table 2 present the performance of VerDA in comparison to all baseline approaches on the CHAIR and POPE benchmark, respectively. The results for OPERA on LLaVA-NeXT-8B are absent due to its excessive memory requirements. Compared with existing hallucination mitigation methods, our method achieves state-of-the-art performance on both CHAIR and POPE benchmarks. Evaluation results on AMBER and MMHal-Bench are presented in the Appendix E.3 for a more comprehensive assessment of hallucination mitigation performance. More qualitative results can be found in Appendix E.4.

Table 1: CHAIR evaluation results on the MSCOCO dataset averaged over 5 random splits. D_1 denotes the Distinct-1 score of generated captions. Higher D_1 indicates more lexically diverse and less repetitive generation. The best and second-best results are highlighted in **bold** and underlined.

Method	Venue	LLaVA-1.5-7B			LLaVA-Next-8B			InstructBLIP-13B		
		$CHAIR_S \downarrow$	$CHAIR_I \downarrow$	$D_1 \uparrow$	$CHAIR_S \downarrow$	$CHAIR_I \downarrow$	$D_1 \uparrow$	$CHAIR_S \downarrow$	$CHAIR_I \downarrow$	$D_1 \uparrow$
Beam	–	$52.96_{\pm 2.41}$	$15.31_{\pm 0.39}$	0.0624	$\underline{27.94}_{\pm 1.73}$	$7.57_{\pm 0.49}$	0.0331	$50.20_{\pm 1.74}$	$13.43_{\pm 0.31}$	0.0592
CODE [19]	NeurIPS 2024	$48.40_{\pm 1.41}$	$14.28_{\pm 0.53}$	0.0692	$30.42_{\pm 1.39}$	$8.03_{\pm 0.39}$	0.0363	$55.92_{\pm 3.28}$	$15.54_{\pm 1.24}$	0.0607
VCD [16]	CVPR 2024	$51.56_{\pm 2.14}$	$15.15_{\pm 0.72}$	0.0652	$32.01_{\pm 1.57}$	$8.24_{\pm 0.91}$	0.0347	$51.80_{\pm 2.02}$	$14.42_{\pm 0.42}$	0.0602
OPERA [18]	CVPR 2024	$42.96_{\pm 1.56}$	$13.20_{\pm 0.33}$	0.0739	–	–	–	$50.56_{\pm 1.94}$	$14.56_{\pm 0.89}$	0.0639
DoLa [17]	ICLR 2024	$51.56_{\pm 2.79}$	$14.79_{\pm 1.03}$	0.0659	$28.92_{\pm 1.45}$	$7.97_{\pm 0.38}$	0.0347	$51.36_{\pm 1.79}$	$14.49_{\pm 0.78}$	0.0601
DropoutDecoding [20]	NeurIPS 2025	$48.56_{\pm 2.17}$	$14.20_{\pm 0.90}$	0.0667	$28.20_{\pm 1.39}$	$7.67_{\pm 0.42}$	0.0348	$49.30_{\pm 1.81}$	$14.23_{\pm 0.75}$	0.0542
VAF [21]	CVPR 2025	$50.00_{\pm 2.70}$	$14.71_{\pm 0.91}$	0.0668	$28.96_{\pm 1.28}$	$8.08_{\pm 0.51}$	0.0348	$49.20_{\pm 1.58}$	$14.47_{\pm 0.68}$	0.0551
SPARC [24]	ICML 2025	$46.02_{\pm 2.42}$	$14.44_{\pm 3.71}$	0.0664	$31.84_{\pm 1.76}$	$8.76_{\pm 0.74}$	0.0345	$50.32_{\pm 2.03}$	$14.43_{\pm 0.92}$	0.0602
VHR [23]	ACL 2025	$\underline{33.40}_{\pm 1.56}$	$\underline{9.97}_{\pm 0.51}$	0.0675	$28.92_{\pm 1.51}$	$\underline{7.17}_{\pm 0.46}$	0.0361	$\underline{42.48}_{\pm 1.68}$	$\underline{11.98}_{\pm 0.87}$	0.0592
AdaIAT [25]	CVPR 2026	$35.41_{\pm 2.47}$	$10.93_{\pm 0.96}$	$\underline{0.0813}$	$28.04_{\pm 1.62}$	$7.97_{\pm 1.44}$	$\underline{0.0372}$	$42.52_{\pm 0.92}$	$13.22_{\pm 0.61}$	**0.0736**
Baseline (Greedy)	–	$49.76_{\pm 2.02}$	$14.21_{\pm 0.96}$	0.0661	$29.16_{\pm 1.27}$	$7.92_{\pm 0.68}$	0.0348	$50.32_{\pm 2.03}$	$14.43_{\pm 0.92}$	0.0602
VerDA (Ours)	–	**28.68**$_{\pm 2.09}$	**9.45**$_{\pm 0.99}$	**0.0917**	**27.52**$_{\pm 1.04}$	**6.86**$_{\pm 0.61}$	**0.0382**	**41.08**$_{\pm 2.92}$	**11.69**$_{\pm 0.83}$	$\underline{0.0644}$

MME and Other Models Evaluations. Table 3 presents the evaluation results of LLaVA-1.5-7B on a subset of the MME benchmark. And our method achieves the best Total score among all methods. This indicates our method enhances multimodal capabilities across diverse evaluation dimensions, rather than being optimized for a single category. Comprehensive results on the full MME benchmark are provided in the Appendix E.6. We further evaluate VerDA on a broader set of LVLMs to examine its generalization ability. As shown in Table 4, it remains effective at mitigating hallucinations across

Table 2: Accuracy and F1 scores on POPE averaged over popular, adversarial, and random splits.

Method	LLaVA-1.5-7B Acc ↑	LLaVA-1.5-7B F1 ↑	LLaVA-Next-8B Acc ↑	LLaVA-Next-8B F1 ↑	InstructBLIP-13B Acc ↑	InstructBLIP-13B F1 ↑
Beam	84.54	85.01	89.27	89.32	81.89	83.87
CODE [19]	83.23	84.60	88.58	89.09	81.41	83.67
VCD [16]	82.76	83.92	88.31	88.67	81.99	83.91
OPERA [18]	84.67	85.47	–	–	76.27	80.35
DoLa [17]	84.03	85.10	89.09	89.46	81.22	83.59
DropoutDecoding [20]	83.97	84.99	88.99	89.38	81.34	83.56
VAF [21]	83.38	84.72	88.53	89.09	77.82	81.57
SPARC [24]	83.84	84.97	89.02	89.41	81.42	83.66
VHR [23]	84.00	85.49	89.04	89.40	82.64	84.21
AdaIAT [25]	83.85	84.99	88.91	89.36	81.45	83.69
Baseline (Greedy)	83.84	84.97	89.02	89.41	81.42	83.66
VerDA (Ours)	**84.71**	**85.55**	**89.39**	**89.71**	**83.11**	**84.63**

Table 3: Performance evaluation on the subset of MME benchmark.

Method	Exist. ↑	Count ↑	Pos. ↑	Color ↑	Total ↑
Beam	195.00	136.67	123.33	145.00	600.00
CODE [19]	195.00	143.33	113.33	148.33	599.99
VCD [16]	190.00	148.33	123.33	143.33	604.99
OPERA [18]	195.00	153.33	128.33	148.33	624.99
DoLa [17]	195.00	148.33	128.33	150.00	621.66
DropoutDecoding [20]	195.00	148.33	128.33	150.00	621.66
VAF [21]	190.00	148.33	133.33	153.33	624.99
VHR [23]	195.00	143.33	113.33	150.00	601.66
SPARC [24]	195.00	148.34	128.33	150.00	621.67
AdaIAT [25]	195.00	148.33	128.33	150.00	621.67
Baseline (Greedy)	190.00	148.33	128.33	148.33	615.00
VerDA (Ours)	195.00	158.33	128.33	150.00	**631.67**

diverse model architectures and parameter scales, even on state-of-the-art models such as Qwen3-VL and InternVL3, as well as larger models.

4.2 Evaluation Results on Generation Quality Benchmarks.

CapMAS Evaluations. Table 5 presents the evaluation results on the CapMAS benchmark, scored by GPT-4o following the original paper [41]. These results demonstrate that VerDA consistently improves performance across all evaluated models on more challenging tasks in recent benchmarks. Notably, the most pronounced gains are observed in factuality, indicating that VerDA effectively mitigates hallucinations and strengthens alignment with visual cues. Improvements in coverage suggest that the model captures a broader set of relevant visual details, as removing redundant tokens enables attention to focus on complementary visual cues. Meanwhile, gains in CLAIR score indicate improved alignment between generated content and visual inputs.

Table 4: Generalization performance of VerDA across other LVLMs on CHAIR and POPE.

Model	$CHAIR_S$ ↓	$CHAIR_I$ ↓	POPE-Acc ↑	POPE-F1 ↑
GLM-4V-9B	39.71	8.98	89.45	89.34
w/ VerDA	**36.06** (+3.65)	**7.94** (+1.04)	**89.69** (+0.24)	**89.59** (+0.25)
Qwen2.5-VL-7B	25.03	8.46	82.95	79.94
w/ VerDA	**19.88** (+5.15)	**7.19** (+1.27)	**83.45** (+0.50)	**80.67** (+0.73)
Qwen3-VL-8B	50.68	9.97	88.71	88.62
w/ VerDA	**41.52** (+9.16)	**9.05** (+0.92)	**89.20** (+0.49)	**89.16** (+0.54)
InternVL2-8B	36.64	8.34	84.78	82.83
w/ VerDA	**29.24** (+7.40)	**7.36** (+0.98)	**84.96** (+0.18)	**82.96** (+0.13)
InternVL3-9B	30.52	7.67	89.85	89.80
w/ VerDA	**28.24** (+2.28)	**7.31** (+0.36)	**89.96** (+0.11)	**90.02** (+0.22)
InternVL3-14B	33.18	8.22	88.93	88.16
w/ VerDA	**29.84** (+3.34)	**7.78** (+0.44)	**89.63** (+0.70)	**89.09** (+0.93)

Table 5: CapMAS evaluation results, scored by GPT-4o, assessing caption quality across CLAIR score, Factuality, Coverage, and the overall average.

Model	CLAIR ↑	Factuality ↑	Coverage ↑	Avg. ↑
LLaVA-1.5-7B	52.98	52.38	48.74	51.37
w/ VerDA	**54.86** (+1.88)	**60.26** (+7.88)	**49.31** (+0.57)	**54.81** (+3.44)
LLaVA-NeXT-8B	64.28	66.13	66.15	65.52
w/ VerDA	**65.87** (+1.59)	**67.01** (+0.88)	**67.01** (+0.86)	**66.63** (+1.11)
InstructBLIP-13B	46.67	43.29	47.96	45.97
w/ VerDA	**47.57** (+0.90)	**44.21** (+0.92)	**49.71** (+1.75)	**47.16** (+1.19)

Table 6: LLaVA-Bench (In-the-Wild) evaluation results, scored by GPT-4o via pairwise response comparison.

Model	Accuracy	Detailedness	Naturalness
LLaVA-1.5-7B	4.157	4.683	**7.412**
w/ VerDA	**4.563**	**4.777**	7.250
LLaVA-NeXT-8B	6.367	6.087	**7.183**
w/ VerDA	**6.433**	**6.267**	6.983
InstructBLIP-13B	5.156	4.733	6.133
w/ VerDA	**5.333**	**4.830**	**6.267**

LLaVA-Bench Evaluations. Table 6 presents the evaluation results of GPT-4o on LLaVA-Bench (In-the-Wild). These results show that VerDA effectively improves the accuracy of model responses, while achieving comparable or even better performance than the baseline in terms of detailedness and naturalness. This indicates that our method enhances the performance on more diverse and challenging tasks while maintaining response quality.

4.3 Ablation Study

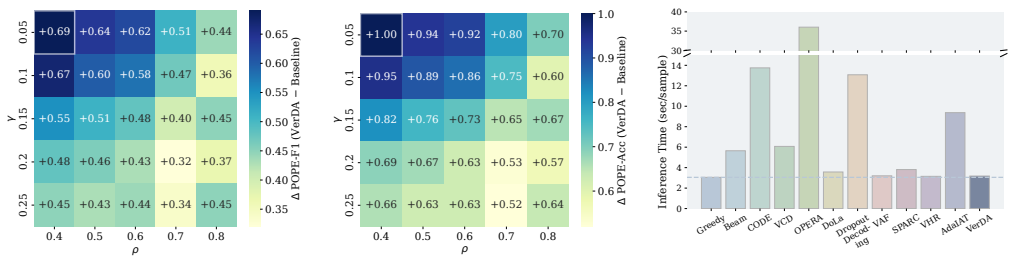


Figure 6: **Left (a) – Middle (b)**: Sensitivity analysis of ρ and γ on POPE, where (a) reports ΔPOPE-F1 and (b) reports ΔPOPE-Acc. Cells show improvements of VerDA over the baseline, with the best results highlighted in white. **Right (c)**: Comparison of inference time for different methods.

Sensitivity of Analysis Hyperparameters. We evaluate the sensitivity of hyperparameters ρ and γ on LLaVA-1.5-7B. As shown in Figure 6 (a) - (b), our method remains stable across different parameter settings and consistently outperforms the baseline under all configurations, demonstrating strong robustness. Additional results on other models are provided in the Appendix E.5.

Additional Time Analysis. Our method only intervenes in the attention during the inference stage, without an additional forward pass. As a result, its computational overhead is negligible. A detailed inference time comparison with baseline methods is shown in Figure 6 (c).

Impact of Head-Adaptive Strategies. Since VerDA selects an adaptive number of visual tokens for each attention head and each sample, we conduct an ablation study to validate the necessity of this strategy. Specifically, we compare VerDA with two variants: *Head-fixed VerDA* uses a fixed number of tokens per head, while the selected regions remain adaptive. *Global-fixed VerDA* uses the same visual regions for all heads, with a fixed number of tokens. To ensure a fair comparison, we align the token budget by setting the number of selected tokens in both variants to match the average number used by VerDA. As shown in Table 7, both variants degrade performance, especially when fixing attended regions, confirming the necessity of head-adaptive token selection.

Table 7: Ablation study on head-adaptive visual token selection strategies.

Model	Method	POPE-Acc ↑	POPE-F1 ↑
LLaVA-1.5-7B	Global-fixed VerDA	83.93	85.04
	Head-fixed VerDA	84.46	85.38
	VerDA	**84.71**	**85.55**
LLaVA-Next-8B	Global-fixed VerDA	89.34	89.66
	Head-fixed VerDA	89.23	89.56
	VerDA	**89.39**	**89.71**
InstructBLIP-13B	Global-fixed VerDA	80.69	83.39
	Head-fixed VerDA	82.79	84.40
	VerDA	**83.11**	**84.63**

Impact of Core Components. In Table 8, we conduct ablation studies on LLaVA-1.5-7B by progressively introducing each component to evaluate the effectiveness of the core components in VerDA. **VD** denotes Head-Adaptive Visual Deredundancy, **DA** denotes Distance-Aware Attenuation, and **VE** denotes Visual Cues Head-Adaptive Enhancement. The results show that each component consistently improves performance, and combining all components achieves the best results, demonstrating their complementary effects. Additional results on other models are provided in Appendix E.6.

Table 8: Ablation study of the core components on CHAIR and POPE. ✓ indicates that the component is included, while ✗ indicates that it is excluded.

Baseline	Components			CHAIR		POPE	
	VD	DA	VE	CHAIRs ↓	CHAIRi ↓	POPE-Acc ↑	POPE-F1 ↑
✓	✗	✗	✗	49.76	14.21	83.84	84.97
✓	✓	✗	✗	44.60	13.02	84.09	85.13
✓	✓	✓	✗	45.60	12.42	84.65	85.42
✓	✓	✓	✓	**28.68**	**9.45**	**84.71**	**85.55**

5 Related Work

Existing decoding strategies [16, 17, 18, 19, 20] mainly intervene at the output or decoding level to reduce reliance on language priors, without directly enhancing internal visual cues. CMI-VLD [42] further purifies visual tokens using a trained module, but aggregates attention scores across heads and applies the same retained visual-token set to all heads. *In contrast, VerDA dynamically preserves sparse visual cues within each head, accounting for their complementarity across heads without additional training.*

Another line of research analyzes hallucination through internal attention behavior, showing a close link between hallucination and visual attention. VAF [21] increases attention to visual tokens while reducing attention to system tokens. SPARC [24] identifies key visual tokens from aggregated attention changes across decoding steps and reinforces them across heads. VAR [22] reallocates attention from visual sink tokens identified using predefined sink dimensions, with the sink token set shared across heads. VHR [23] selects vision-sensitive heads and enhances their attention. AdaIAT [25] considers head differences but relies on external statistics and focuses on text-side attention. Modular Attribution and Intervention [43] identifies hallucination-related heads through dataset-level attribution and suppresses text-token attention through threshold-triggered zeroing. Besides, Few Attention Heads [44] reveals head-level sparsity in visual grounding, rather than visual-token sparsity within each head. Although these studies improve visual grounding or mitigate hallucinations, they do not jointly consider the sparsity and complementarity of head-wise visual cues and their gradual weakening during decoding. *In contrast, VerDA dynamically identifies visual cues within each head and reinforces them through visual de-redundancy and distance-aware text attenuation.* Detailed distinctions are provided in Appendix F.

6 Conclusion

In this work, we revisit hallucination in LVLMs from the perspective of attention head heterogeneity and visual cues. We observe that visual cues are sparse within each head, while different heads capture complementary visual cues, and that the proportion of these cues within overall visual attention declines during generation. Meanwhile, the model's ability to utilize long-range textual context also weakens, jointly contributing to hallucinations. Building on these findings, we propose VerDA, a training-free method to mitigate hallucinations. Specifically, VerDA performs head-adaptive visual de-redundancy to remove low-value visual tokens, applies distance-aware text attenuation to suppress over-reliance on local context, and reallocates the released attention to visual cues to reinforce key visual grounding. Extensive experiments across multiple LVLMs and benchmarks demonstrate that VerDA outperforms existing methods in mitigating hallucinations, with negligible inference overhead.

Limitations

Our analysis and mitigation strategy primarily focus on the attention mechanisms in LVLMs. While these mechanisms play a critical role in hallucinations, other architectural components, such as the combination methods and interaction effects between attention control and different decoding strategies, have not been systematically explored in this work. Moreover, despite the strong hallucination mitigation performance of VerDA, like many existing methods, may lead to more conservative responses in certain models. Future research could further explore the synergy between different mechanisms and develop more comprehensive interventions at the model level, beyond attention-based modulation.

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A Causal Validation via Masking

To complement the retention-based intervention in Figure 2 (d), we further conduct a masking-based intervention to examine the necessity of the selected visual cues. While the retention experiment evaluates whether the selected tokens are sufficient to preserve reliable grounding under a constrained visual-token budget, the masking experiment asks whether removing these tokens causes a larger performance degradation than removing other tokens. Specifically, we mask visual tokens selected by four strategies: Per-Head Top-k, Global Top-k, Per-Head Random-k, and Per-Head Bottom-k. The masking ratio is fixed to $k = 0.1$ for all strategies to ensure a fair comparison.

Table 9: Masking-based necessity analysis of head-wise visual cues on POPE [9] benchmark. Positive values indicate improvement over the baseline, while negative values indicate degradation. The largest degradation is highlighted in **bold**.

Metric	**w/o Per-Head Top-k**	**w/o Global Top-k**	**w/o Per-Head Random-k**	**w/o Per-Head Bottom-k**
ΔAccuracy	**-0.43**	-0.13	+0.02	+0.08
ΔF1-score	**-0.21**	-0.07	+0.01	+0.05

As shown in Table 9, masking Per-Head Top-k tokens leads to the largest degradation in both Accuracy and F1-score. In contrast, masking Per-Head Random-k or Per-Head Bottom-k tokens does not degrade performance, suggesting that these tokens are less critical and may contain redundant or distracting information. Compared with Global Top-k, the stronger degradation caused by masking PH Top-k further indicates that visual cues are distributed in a head-specific manner and cannot be fully captured by globally aggregated attention. Together with the retention-based intervention in Figure 2 (d), these results provide complementary causal support for using head-wise high attention visual tokens as an operational proxy for visual cues.

B VerDA Algorithm

Algorithm 1 outlines VerDA. For each selected layer and attention head, VerDA performs three steps: (i) head-adaptive visual de-redundancy to select and retain key visual cues while removing redundant tokens; (ii) distance-aware text attenuation to suppress local textual bias and release attention; and (iii) visual cues reinforcement, which reallocates the released attention to enhance visual cues.

Algorithm 1 VerDA

Input: visual index set $\mathcal{V}$, text index set $\mathcal{T}$, coverage ratio ρ, minimum retention ratio γ, reinforced layer set L_r.
for each layer $l \in L_r$ **do**
  for each attention head h **do**
    Select visual cues tokens $\mathcal{I}_h$ from $\mathcal{V}$ by Eq. (8);
    De-redundant visual tokens by Eq. (9);
    Compute released visual attention by $R_{t,\mathcal{V}}^{(l,h)} = \sum_{i \in \mathcal{V} \setminus \mathcal{I}_{l,h}^*} a_{t,i}^{(l,h)}$;
    for each text token $j \in \mathcal{T}$ **do**
      Attenuate text attention based on distance-aware by Eq. (10) and Eq. (11);
    end for
    Compute released text attention by $R_{t,\mathcal{T}}^{l,h} = \sum_{j \in \mathcal{T}} \left(a_{t,j}^{(l,h)} - \hat{a}_{t,j}^{(l,h)} \right)$;
    for each token $i \in \mathcal{I}_{l,h}^*$ **do**
      Enhance visual cues by Eq. (12);
    end for
  end for
end for

C Models and Baselines

Models. To demonstrate the broad applicability of VerDA across LVLM architectures and scales, we evaluate it on a diverse set of recent models, including LLaVA-1.5-7B [2], which is built on the

vicuna backbone, and LLaVA-NeXT-8B [33], which adopts the more recent llama3 backbone, both following CLIP-based instruction-tuned paradigms; InstructBLIP-13B [3], which adopts a Q-Former for visual-linguistic alignment; Qwen2.5-VL-7B [34] and Qwen3-VL-8B [4], which are unified multimodal foundation models; InternVL2-8B and InternVL3-9B/14B [35, 36, 5], which emphasize strong visual encoding and scaling; and GLM-4V-9B [37], a multimodal extension of the GLM series.

Baselines. To verify the effectiveness of VerDA in mitigating hallucinations, we evaluate it against two baseline decoding methods, five representative strategies, and four state-of-the-art attention intervention methods. (1) Baseline decoding methods: (i) Greedy decoding selects the token with the highest probability at each generation step; (ii) Beam search maintains the top-k most likely candidate sequences at each step to search for a high-quality output. (2) Decoding strategies: (i) CODE [19] performs contrastive decoding using self-generated descriptions; (ii) VCD [16] contrasts the output distributions conditioned on the original and perturbed images; (iii) OPERA [18] alleviates hallucination by penalizing over-trusted attention patterns and reallocating attention through a retrospection mechanism; (iv) DoLa [17] improves factuality by contrasting the output distributions from different transformer layers during decoding; (v) DropoutDecoding [20] enhances generation reliability by applying uncertainty-guided dropout during decoding. (3) Attention intervention methods: (i) VAF [21] enhances attention to all visual tokens while suppressing attention to system tokens; (ii) SPARC [24] identifies and reinforces key visual tokens based on attention differences across decoding steps; (iii) VHR [23] uses vision-aware head divergence to identify and enhance visually sensitive attention heads; (iv) AdaIAT [25] adaptively enhances attention to generated text to improve contextual grounding and reduce hallucinations.

D Benchmarks and Evaluation Metrics

To comprehensively evaluate the effectiveness of our method, we conducted extensive experiments on the following benchmarks and metrics.

CHAIR. CHAIR [8] evaluates object hallucinations in LVLMs through image captioning. We randomly select 500 images from the val2014 split of the MSCOCO [45] dataset and use the prompt "Please describe the image in detail." for LVLMs to generate image captions. The CHAIR metric evaluates object hallucinations by comparing the objects mentioned in the generated captions with the ground-truth object annotations. It includes the sentence-level hallucination evaluation (CHAIR_S) and the object-level hallucination evaluation (CHAIR_I), defined as follows:

$$\begin{aligned} \text{CHAIR}_S &= \frac{|\{\text{captions with hallucinated objects}\}|}{|\{\text{all captions}\}|}, \\ \text{CHAIR}_I &= \frac{|\{\text{hallucinated objects}\}|}{|\{\text{all mentioned objects}\}|}. \end{aligned} \tag{13}$$

To further evaluate the semantic diversity of generated captions, we additionally adopt Distinct-1 (D_1) [46], a widely used metric for measuring lexical diversity. D_1 is defined as the ratio of the number of unique unigrams to the total number of generated tokens. In our implementation, we compute D_1 at the corpus level by aggregating tokens across all generated captions. Specifically, each caption is first normalized by converting to lowercase and removing punctuation, followed by whitespace tokenization. The Distinct-1 score is then computed as:

$$D_1 = \frac{|\{\text{unique tokens}\}|}{|\{\text{all tokens}\}|}. \tag{14}$$

A higher D_1 score indicates richer vocabulary usage and reduced repetition in generated descriptions. This metric complements CHAIR by providing an additional perspective on generation quality, capturing the trade-off between factual correctness and linguistic diversity.

POPE. POPE [9] is a popular benchmark for assessing object hallucinations in LVLMs, based on binary visual question answering (VQA). This benchmark queries LVLMs with yes-or-no questions regarding the presence of specific objects, i.e., "Is there a <object> in the image?". The evaluation includes three subsets—*random*, *popular*, and *adversarial*—based on different strategies to generate

negative samples. Each image is associated with 6 questions, resulting in a total of 3,000 test instances. The metrics include Accuracy, Precision, Recall, and F1 scores, with the results averaged across all three subsets.

MME. MME [40] is a comprehensive benchmark for LVLMs, assessing both their perception and reasoning capabilities. It covers 14 different tasks, and we selected 4 of these tasks for hallucination assessment, i.e., *existence* and *count* for object-level hallucinations, and *position* and *color* for attribute-level hallucinations. Each subset contains 30 images, with two yes-or-no questions per image, resulting in 60 questions per subset. Following the official protocol, performance is measured using both accuracy and accuracy+, where accuracy is calculated for individual questions, and accuracy+ measures the proportion of images where both questions are answered correctly.

CapMAS. CapMAS [41] is a recently proposed benchmark for hyper-detailed image captioning, constructed from a curated set of 1,000 images, including 600 images from DOCCI [47] and 400 images from the IIW-400 [48] dataset, which contains images paired with highly detailed, hallucination-free captions. It adopts *Factuality* and *Coverage* as fine-grained evaluation metrics to assess the correctness and completeness of generated descriptions, respectively. Specifically, *Factuality* measures the degree of truthfulness and the absence of hallucinated content, while *Coverage* evaluates how comprehensively the visual content is described. Both metrics are computed using the CLAIR framework [49], which further provides an overall *CLAIR* score to assess whether the generated captions are supported by the image. Following the official protocol, we collect model responses from five different prompts and use Meta-LLaMA-3-8B-Instruct [2] to merge them into a single caption for evaluation. The final scores are obtained using GPT-4o as the evaluator.

LLaVA-Bench LLaVA-Bench (In-the-Wild) [2] evaluates the performance of LVLMs on more challenging tasks. It consists of 24 images covering diverse and challenging scenarios, including complex scenes, memes, paintings, and sketches, along with 60 challenging questions. Due to the open-ended nature of the task, we use the GPT-4o model to evaluate the responses of LVLMs in terms of accuracy, detailedness, and naturalness.

AMBER AMBER [38] is an LLM-free benchmark designed to evaluate hallucinations in LVLMs across both generative and discriminative tasks. It consists of 1,004 carefully curated images with fine-grained annotations covering three types of hallucinations: object existence, attributes, and relations. For generative evaluation, models are prompted to describe the given images, and their responses are assessed using four metrics: *CHAIR*, *Cover*, *Hal*, and *Cog*. Specifically, *CHAIR* measures the proportion of hallucinated objects, *Cover* evaluates the coverage of ground-truth objects, *Hal* measures the proportion of responses containing hallucinations, and *Cog* quantifies hallucinations involving plausible but absent objects. For discriminative evaluation, models answer binary questions with performance measured using accuracy, precision, recall, and F1 score. The benchmark also provides an overall *AMBER Score* that combines generative CHAIR and discriminative F1 scores.

MMHal-Bench MMHal-Bench [39] is designed to assess the factual consistency of open-ended responses generated by LVLMs. It consists of 96 challenging image-question pairs constructed from the OpenImages dataset, spanning 8 question categories and 12 object topics. Due to the open-ended nature of the task, we use the GPT-4o model to evaluate the responses of LVLMs. Each response receives a score ranging from 0 to 6, considering both informativeness and factual consistency, where higher scores indicate better performance.

E More Experiment Results and Analyses

E.1 Layer Configuration and Ablation Analysis

Layer Configuration. Unless otherwise specified, all additional analyses follow the same model-specific settings as VerDA in the main experiments. We apply VerDA to a fixed subset of transformer layers for each model series and use the same configuration across all benchmarks. The layer sets are not tuned separately for individual benchmarks. This subset-based intervention keeps the

[2] Meta-LLaMA-3-8B-Instruct: https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct

method lightweight and avoids excessive perturbation of the original attention dynamics. Our layer configuration considers two groups of layers. First, motivated by prior work [50], which suggests that shallow layers are important for early visual-text alignment, we include the first two layers for several model families where shallow-layer intervention shows stable gains, including LLaVA-1.5, LLaVA-NeXT, and InternVL series. Second, we select a sparse set of middle-to-deep layers, where our layer-wise analysis shows more pronounced and stable visual discriminability and cross-head complementarity. As shown in Figure 7, these layers exhibit stronger head-wise differences in visual token usage and broader union coverage across heads, suggesting that head-specific visual references are more clearly separated in these layers. We therefore apply VerDA to these middle-to-deep layers in an interval-based manner to avoid overly dense intervention. Different LVLM families differ in how visual representations are encoded, compressed, and organized before entering the language decoder. Therefore, we use a fixed layer configuration for each model series, while keeping the configuration unchanged across all benchmarks. For Qwen-VL and GLM-4V, we focus on middle-to-deep layers where head-wise visual complementarity is more stable. For InstructBLIP, whose visual information is compressed by a Q-Former before being passed to the language model, we use a smaller coverage ratio ρ and intervene only on a compact set of layers. Concretely, for LLaVA-1.5-7B and LLaVA-NeXT-8B, we apply VerDA to layers $\{0, 1, 11\}$ and $\{0, 1, 15, 17, 19, 21, 23, 25, 27\}$, respectively. For the Qwen-VL series, we apply VerDA to selected intermediate and upper layers. Specifically, we use layers $13, 15, 17, 19, 21, 23$ for Qwen2.5-VL-7B, and $13, 15, 17, 19, 21, 23, 25$ for Qwen3-VL-8B. For the InternVL series, we use layers $\{0, 1, 13, 15, 17, 19, 21, 23, 25\}$. For InstructBLIP-13B, we use layers $\{7, 9\}$. For GLM-4V-9B, we use layers $\{22, 24, 26, 28, 30, 32, 34\}$.

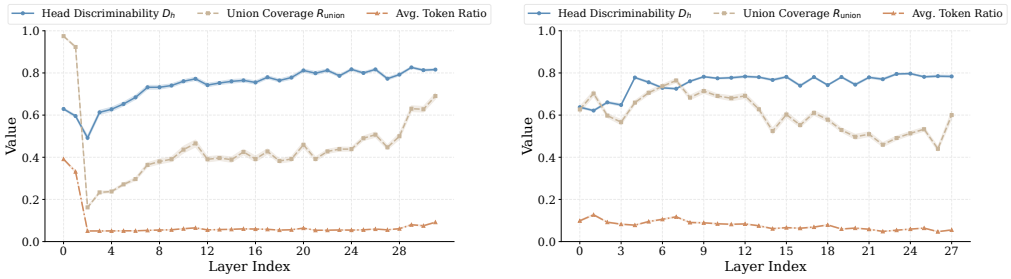


Figure 7: Layer-wise analysis of head discriminability D_h, union coverage ratio R_{union}, and average visual token ratio $|\mathcal{I}_{l,h}|/|V|$ under the coverage constraint $\rho = 0.6$. A higher D_h indicates more diverse visual regions attended across heads, while a larger R_{union} implies stronger cross-head complementarity and broader visual coverage. **Left (a):** LLaVA-1.5-7B. **Right (b):** Qwen2.5-VL-7B.

Ablation Analysis. We conduct layer-wise ablation studies on LLaVA-1.5-7B to validate the effectiveness of our layer selection strategy. Specifically, we compare three configurations: (i) applying VerDA only to middle layers $\{11\}$, (ii) applying VerDA only to layers $\{0, 1\}$, and (iii) applying VerDA to both shallow and middle layers (our full setting). Table 10 shows that applying VerDA to shallow layers and middle-to-deep layers both effectively mitigates hallucinations, with their combination yielding the best overall performance. Specifically, intervening only on layer $\{11\}$ yields modest improvements, whereas extending VerDA to layers $\{0,1\}$ yields a substantial reduction on the CHAIR benchmark. Further incorporating middle-to-deep layers consistently achieves the strongest performance across benchmarks. These results suggest that VerDA mitigates hallucinations through distinct mechanisms at different depths. Further analysis of the layer-specific mechanisms in LVLMs remains an important direction for future work.

Table 10: Ablation on layer configurations of VerDA.

Method	CHAIRs ↓	CHAIRi ↓	POPE-Acc ↑	POPE-F1 ↑	Exist. ↑	Count ↑	Pos. ↑	Color ↑	Total ↑
LLaVA-1.5-7B	49.76	14.21	83.84	84.97	190.00	148.33	128.34	148.33	615.00
w/ VerDA on middle layers	47.21	13.91	83.95	85.04	195.00	158.33	128.34	148.33	630.00
w/ VerDA on shallow layers	29.36	10.08	84.63	85.49	195.00	153.33	128.34	150.00	626.67
w/ VerDA on both	**28.68**	**9.45**	**84.71**	**85.55**	**195.00**	**158.33**	**128.34**	**150.00**	**631.67**

E.2 Evaluation Results on the Full MME Benchmark

We have expanded the original Table 3 in our paper to report the complete results across all perception and cognition dimensions of MME, together with the overall MME score. As shown in Table 11,

Table 11: Complete evaluation results on all 14 MME subtasks. The best results are highlighted in **bold**. Overall denotes the sum of scores across all 14 subtasks.

Method	Exist. ↑	Count ↑	Pos. ↑	Color ↑	Poster ↑	Celeb. ↑	Scene ↑	Land. ↑	Art ↑	OCR ↑	Comm. ↑	Num. ↑	Trans. ↑	Code ↑	Overall ↑
Beam Search	**195.00**	136.67	123.33	145.00	142.52	135.00	158.75	160.25	116.25	132.50	**122.86**	47.50	**82.50**	50.00	1748.13
CODE [19]	**195.00**	143.33	113.33	148.33	144.56	143.82	157.75	160.00	**121.75**	125.00	115.71	42.50	62.50	60.00	1733.58
VCD [16]	190.00	148.33	123.33	143.33	141.50	**148.53**	156.50	121.25	121.25	130.00	112.14	45.00	55.00	65.00	1701.16
OPERA [18]	**195.00**	153.33	128.33	148.33	142.52	136.76	156.50	159.75	120.25	140.00	107.86	45.00	55.00	65.00	1753.63
DoLa [17]	**195.00**	148.33	128.33	150.00	142.52	137.94	155.75	155.75	118.75	140.00	112.14	47.50	55.00	70.00	1757.01
DropDecoding [20]	**195.00**	148.33	128.33	150.00	143.54	137.94	155.75	161.25	117.50	**147.50**	112.14	47.50	55.00	**75.00**	1774.78
VAF [21]	190.00	148.33	**133.33**	**153.33**	144.56	138.53	155.25	158.00	119.00	**147.50**	118.57	**50.00**	55.00	70.00	1781.40
VHR [23]	**195.00**	143.33	113.33	150.00	134.69	121.18	**161.75**	155.50	110.50	115.00	100.71	40.00	**82.50**	47.50	1670.99
SPARC [24]	**195.00**	148.34	128.33	150.00	143.54	137.06	157.25	**162.00**	118.25	**147.50**	112.14	45.00	55.00	65.00	1764.41
AdaIAT [25]	**195.00**	148.33	128.33	150.00	143.54	137.06	156.50	161.25	117.50	**147.50**	112.42	47.50	55.00	**75.00**	1774.93
Baseline (Greedy)	190.00	148.33	128.33	148.33	142.52	137.06	159.50	159.50	118.50	132.50	**122.86**	47.50	55.00	50.00	1739.93
VerDA (Ours)	**195.00**	**158.33**	128.33	150.00	**145.59**	133.24	156.50	160.75	116.25	**147.50**	118.57	47.50	57.50	72.50	**1787.56** (+47.63)

VerDA improves the model's general performance, outperforming greedy decoding by **47.63 points in the overall score**.

E.3 Evaluation Results on the AMBER and MMHall-Bench

To further evaluate the effectiveness of VerDA in mitigating multimodal hallucinations, we conduct additional experiments on AMBER [38] and MMHal-Bench [39] using Qwen3-VL-8B and LLaVA-NeXT-8B as backbone models. We compare VerDA with greedy decoding and two representative hallucination mitigation methods, VHR [23] and AdaIAT [25]. The evaluation results are presented in Tables 12 and 13, respectively.

Table 12: Evaluation results on AMBER with Qwen3-VL-8B and LLaVA-NeXT-8B. CHAIR, Cover, Hal, and Cog are metrics for the generative task, while Acc., Prec., Rec., and F1 are metrics for the discriminative task. The best results under each backbone are highlighted in **bold**.

Backbone	Method	CHAIR ↓	Cover ↑	Hal ↓	Cog ↓	Acc. ↑	Prec. ↑	Rec. ↑	F1 ↑	AMBER Score ↑
Qwen3-VL-8B	Greedy	7.49	70.29	60.47	**3.38**	88.37	**91.17**	91.28	91.22	91.87
	VHR [23]	7.23	**75.45**	58.12	3.74	88.12	90.42	91.78	91.09	91.93
	AdaIAT [25]	**6.73**	71.35	55.76	3.65	88.37	**91.17**	91.28	91.22	92.25
	VerDA (Ours)	6.98	69.21	**54.12**	3.65	**88.66**	90.87	**92.13**	**91.49**	**92.26**
LLaVA-NeXT-8B	Greedy	9.95	62.53	55.78	4.14	87.33	**91.12**	89.63	90.36	90.20
	VHR [23]	8.84	**63.81**	53.29	4.39	87.44	90.76	90.25	90.50	90.83
	AdaIAT [25]	11.49	54.26	49.90	4.00	87.33	**91.12**	89.63	90.36	89.44
	VerDA (Ours)	**7.74**	62.51	**48.71**	**3.45**	**87.72**	90.90	**90.54**	**90.71**	**91.48**

As shown in Table 12, VerDA achieves the highest AMBER Scores of 92.26 and 91.48 on Qwen3-VL-8B and LLaVA-NeXT-8B, respectively, outperforming greedy decoding by 0.39 and 1.28 points.

Table 13: Evaluation results on MMHal-Bench with Qwen3-VL-8B and LLaVA-NeXT-8B, evaluated by GPT-4o. Avg. Score denotes the overall MMHal-Bench score on a 0–6 scale, and Hall. denotes the hallucination rate (%). The best results under each backbone are highlighted in **bold**.

Backbone	Method	Attr. ↑	Adv. ↑	Comp. ↑	Count ↑	Rel. ↑	Env. ↑	Hol. ↑	Other ↑	Avg. Score ↑	Hall. ↓
Qwen3-VL-8B	Greedy	3.00	2.00	2.17	2.50	3.60	**3.67**	**3.20**	2.33	2.81	64.71
	VHR [23]	1.83	**2.67**	**3.17**	2.67	2.50	2.83	2.60	**3.29**	2.70	71.15
	AdaIAT [25]	3.00	2.17	2.17	**3.00**	**4.20**	2.83	2.20	2.57	2.77	63.46
	VerDA (Ours)	**3.17**	2.00	2.33	2.50	3.80	3.50	**3.20**	3.00	**2.94**	**61.54**
LLaVA-NeXT-8B	Greedy	3.33	3.17	2.58	1.75	2.17	3.67	**1.50**	2.25	2.55	60.40
	VHR [23]	3.42	2.33	2.42	1.17	**2.67**	3.50	1.33	2.17	2.38	70.83
	AdaIAT [25]	**3.67**	2.25	**3.33**	1.50	2.33	3.67	1.17	2.33	2.53	60.42
	VerDA (Ours)	3.37	**3.25**	2.83	**2.50**	2.33	**4.25**	1.33	**3.17**	**2.88**	**57.29**

As shown in Table 13, VerDA consistently achieves the highest average scores and the lowest hallucination rates across both backbone models. Specifically, VerDA improves the average score from 2.81 to 2.94 on Qwen3-VL-8B and from 2.55 to 2.88 on LLaVA-NeXT-8B compared with greedy decoding. Meanwhile, the hallucination rate decreases from 64.71% to 61.54% on Qwen3-VL-8B and from 60.40% to 57.29% on LLaVA-NeXT-8B, corresponding to reductions of 3.17 and 3.11 percentage points, respectively. These results further demonstrate the effectiveness of VerDA in reducing hallucinations while improving the overall quality of open-ended multimodal responses.

E.4 Qualitative Results

As shown in Figure 8, 9 and 10, we present more examples that illustrate the effect of VerDA in eliminating hallucinated objects. After incorporating VerDA, the descriptions generated by the LVLMs faithfully align with the content of the images, while preserving the richness of the descriptions.

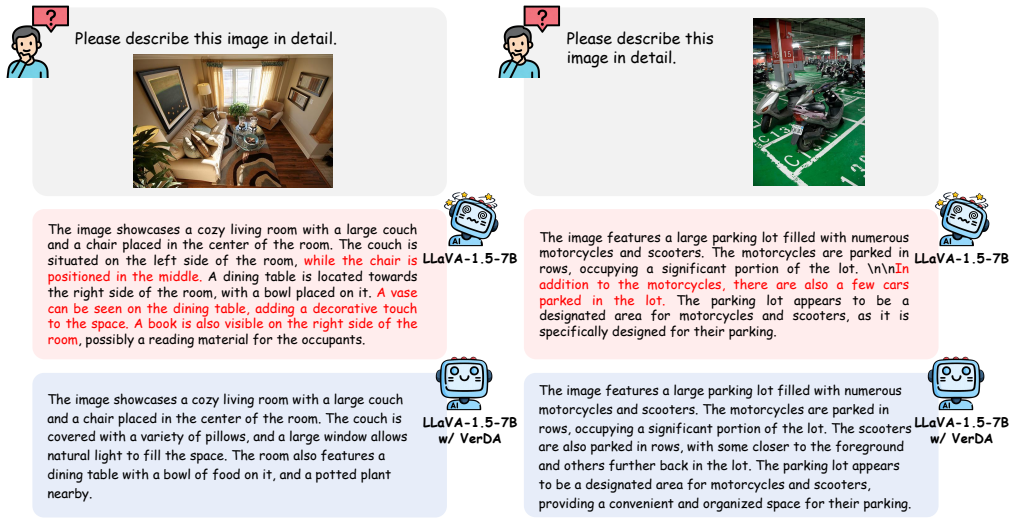


Figure 8: Examples of VerDA mitigating the hallucinated content in LLaVA-1.5-7B.

E.5 Hyperparameter Sensitivity on More Models

We further evaluate the sensitivity of hyperparameters across different model architectures to assess the robustness of VerDA. As shown in Figure 11, our method remains stable across a wide range of parameter settings and consistently outperforms the baseline under all configurations, demonstrating strong robustness.

E.6 Core Components Ablation on More Models

We further evaluate the effectiveness of each component across different model architectures to verify the generality of VerDA. In Table 14, we conduct component ablation studies on LLaVA-NeXT-8B and InstructBLIP-13B by progressively introducing each component to evaluate the effectiveness of the core modules in VerDA. **VD** denotes Head-Adaptive Visual De-redundancy, **DA** denotes Distance-Aware Attenuation, and **VE** denotes Visual Cues Head-Adaptive Enhancement. The results show that each component consistently improves performance across different architectures, and combining all components achieves the best results, demonstrating their complementary effects. This confirms the generality of VerDA across diverse model architectures.

Table 14: Ablation of the core components on different architectures. **Left:** LLaVA-Next-8B, **Right:** InstructBLIP-13B. ✓ indicates that the component is included, while ✗ indicates that it is excluded.

Baseline	Components			CHAIR		POPE	
	VD	DA	VE	CHAIRs ↓	CHAIRi ↓	POPE-Acc ↑	POPE-F1 ↑
✓	✗	✗	✗	29.16	7.92	89.02	89.41
✓	✓	✗	✗	28.80	7.52	89.07	89.48
✓	✓	✓	✗	27.89	7.19	89.09	89.49
✓	✓	✓	✓	**27.52**	**6.86**	**89.39**	**89.71**

Baseline	Components			CHAIR		POPE	
	VD	DA	VE	CHAIRs ↓	CHAIRi ↓	POPE-Acc ↑	POPE-F1 ↑
✓	✗	✗	✗	50.32	14.43	81.42	83.66
✓	✓	✗	✗	47.20	13.02	82.09	84.12
✓	✓	✓	✗	43.00	12.79	81.91	83.96
✓	✓	✓	✓	**39.04**	**11.67**	**83.11**	**84.63**

E.7 Statistical Significance Analysis

To assess whether the performance improvements of our method are statistically significant, we conduct a comprehensive significance analysis on the CHAIR benchmark using the LLaVA-1.5-7B. Specifically, we perform paired t-tests between VerDA and each baseline across five random splits,

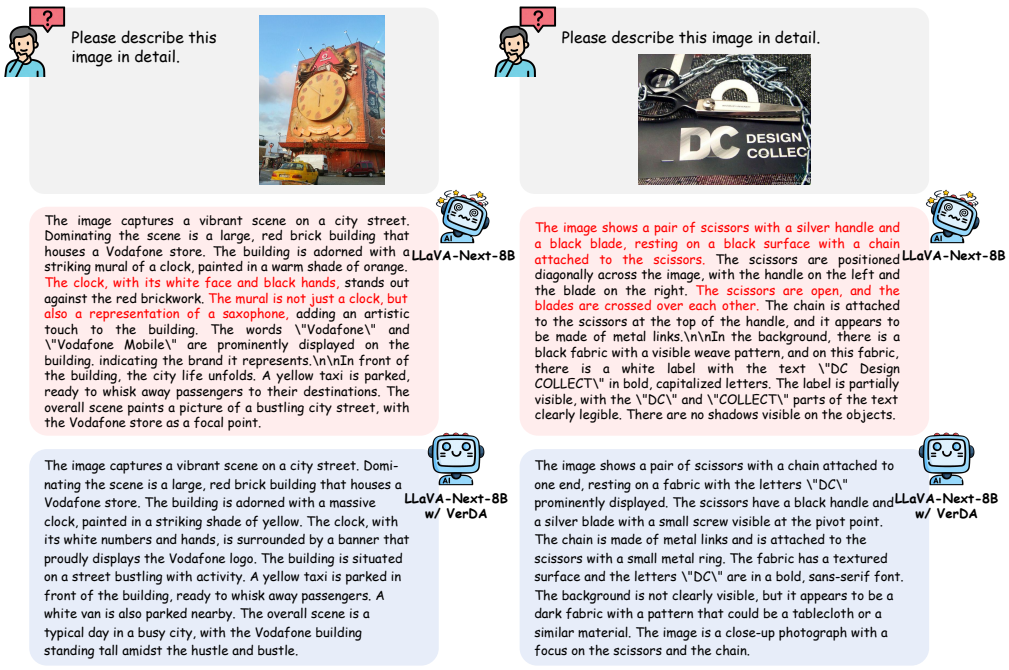


Figure 9: Examples of VerDA mitigating the hallucinated content in LLaVA-Next-8B.

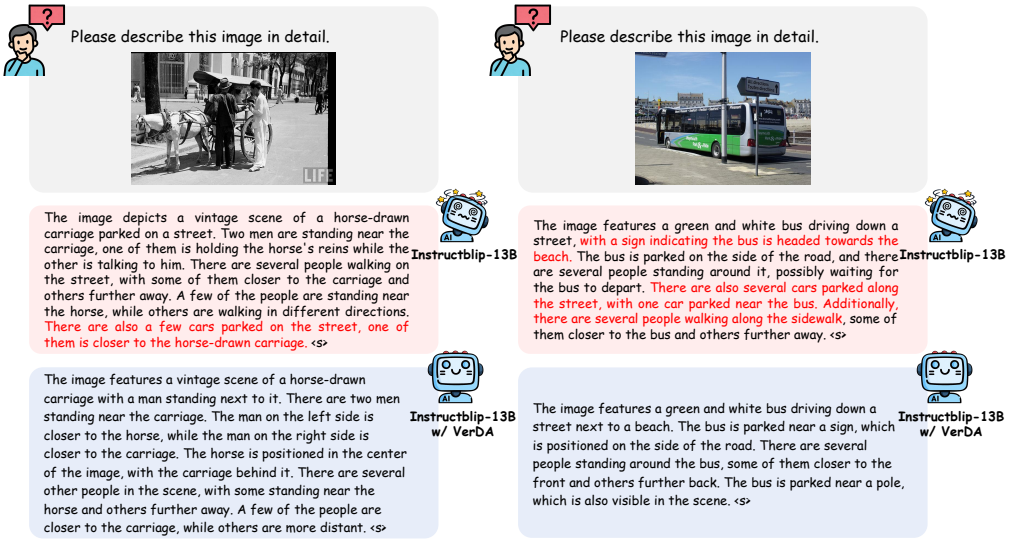


Figure 10: Examples of VerDA mitigating the hallucinated content in InstructBLIP-13B.

evaluating both CHAIR_S and CHAIR_I metrics. We adopt a significance level of $\alpha = 0.05$ and report two-sided p-values. For better interpretability, we visualize statistical significance using $-\log_{10}(p)$, where larger values indicate stronger significance. As shown in the Figure 12, VerDA consistently achieves statistically significant improvements over all baselines on both CHAIR_S and CHAIR_I. In particular, most comparisons reach a highly significant level ($p < 0.01$), demonstrating that the performance gains of VerDA are not only numerically superior but also statistically robust.

E.8 Object Grounding and Visual-Sink Robustness Analysis

To further investigate whether VerDA enhances attention to semantically relevant visual regions rather than reinforcing irrelevant visual sinks, we conduct additional analyses on object grounding

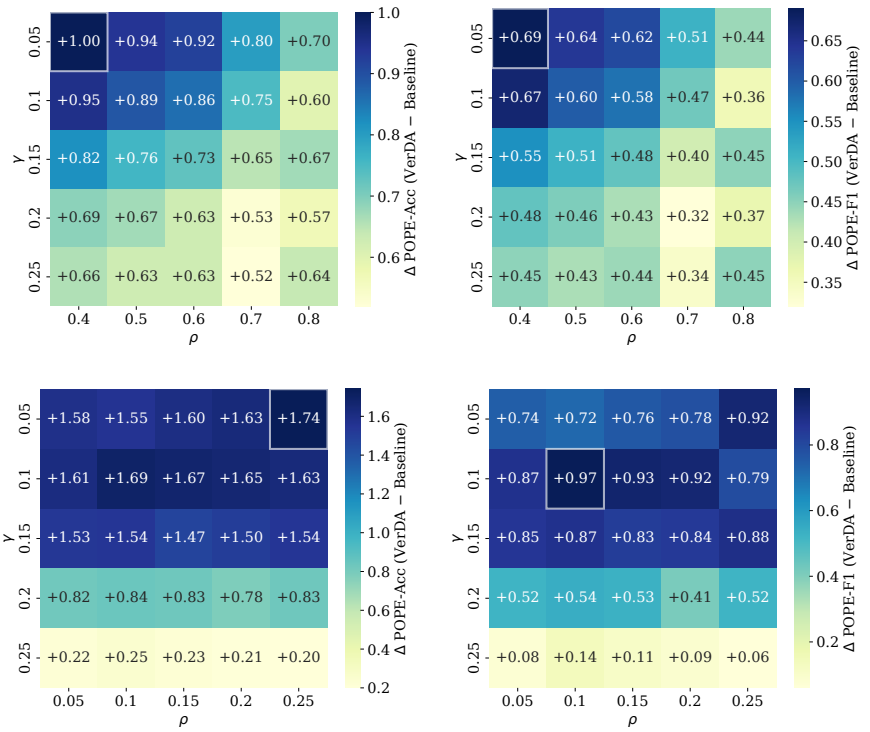


Figure 11: Sensitivity analysis of hyperparameters ρ and γ on POPE. **Upper Left (a):** ΔPOPE-Acc on LLaVA-NeXT-8B. **Upper Right (b):** ΔPOPE-F1 on LLaVA-NeXT-8B. **Lower Left (c):** ΔPOPE-Acc on InstructBLIP-13B. **Lower Right (d):** ΔPOPE-F1 on InstructBLIP-13B. All values are reported as improvements over the baseline, with the best results highlighted in white.

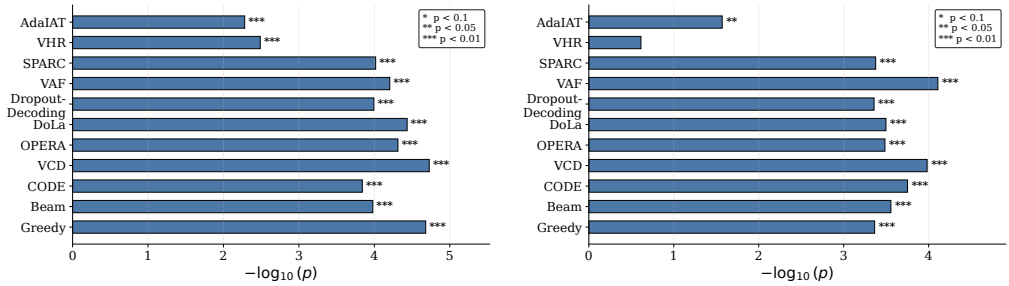


Figure 12: Statistical significance results on the CHAIR benchmark. Each bar shows the $-\log_{10}(p)$ value from a paired t-test comparing VerDA with a baseline, where larger values indicate stronger significance. Asterisks denote significance levels ($*\, p < 0.1$, $**\, p < 0.05$, $***\, p < 0.01$). **Left (a):** Statistical significance analysis on CHAIR$_S$. **Right (b):** Statistical significance analysis on CHAIR$_I$.

selectivity and visual-sink robustness. We randomly sample 500 question-image pairs from the three POPE subsets with a fixed random seed of 0, restricting the samples to those with valid COCO instance-segmentation masks for the queried object categories. We evaluate object grounding selectivity on both LLaVA-NeXT-8B and LLaVA-1.5-7B, while visual-sink robustness is examined on LLaVA-1.5-7B, following the official sink-detection configuration of [23].

Object Grounding Selectivity. To quantify whether VerDA preferentially enhances attention to semantically relevant object regions, we introduce *Attention Contrast*, defined as the ratio of the mean attention per ground-truth (GT) object-region token to that per background token:

$$\text{Attention Contrast} = \frac{\text{MeanAttn(GT)}}{\text{MeanAttn(Background)}}. \tag{15}$$

Here, GT regions are identified using COCO instance-segmentation annotations, and background tokens refer to visual tokens outside the queried object regions. A higher Attention Contrast indicates stronger attention allocation to the target object relative to non-target regions.

VerDA consistently improves Attention Contrast across both backbones. Specifically, on LLaVA-NeXT-8B, Attention Contrast increases from 1.1764 to 1.8584, yielding an absolute improvement of 0.6820. Similarly, on LLaVA-1.5-7B, it increases from 0.8815 to 1.0821, corresponding to an improvement of 0.2006. These results suggest that VerDA preferentially strengthens attention to semantically relevant object regions, rather than indiscriminately amplifying attention to arbitrary visual patches.

Visual-Sink Robustness. We further investigate whether VerDA increases attention concentration on visual sink tokens. Following [**?**], we identify the visual-sink token set $\mathcal{S}^{(l)}$ at layer l using the official detection configuration for LLaVA-1.5-7B. Let $\mathcal{V}$ denote the set of 576 visual tokens, and let $a_{t,j}^{(l,h)}$ represent the attention weight assigned to visual token j at query position t, layer l, and attention head h. We define *Sink Attention Share* as the fraction of total visual attention allocated to sink tokens:

$$\text{Sink Attention Share} = \frac{\sum_{h=1}^{H} \sum_{j \in \mathcal{S}^{(l)}} a_{t,j}^{(l,h)}}{\sum_{h=1}^{H} \sum_{j \in \mathcal{V}} a_{t,j}^{(l,h)}}. \tag{16}$$

A lower Sink Attention Share indicates that a smaller fraction of the visual-attention budget is allocated to sink tokens. Averaged across the evaluated VerDA layers and all 500 samples, VerDA substantially reduces Sink Attention Share from 15.282% to 5.385%, corresponding to a relative reduction of 64.8%. This result indicates that VerDA reduces, rather than reinforces, attention concentration on visual sinks.

F Comparison with Existing Training-free Methods

To further clarify the distinctions between VerDA and existing training-free hallucination mitigation methods, we summarize their core intervention mechanisms in Table 15. Rather than comparing only whether a method requires additional training, we focus on where the intervention is applied, what information is primarily targeted, and whether the method explicitly models head-wise behavior, visual forgetting, and text-side context utilization. As shown in the Table 15, most prior methods intervene at the logits level or rely on aggregated attention patterns, while only partially considering the internal mechanisms that cause visual cues to be weakened during generation. In contrast, VerDA directly operates on attention, reinforces head-wise visual cues, models visual forgetting, and incorporates distance-aware text-side attenuation without relying on external statistics.

Table 15: Comparison between VerDA and representative training-free hallucination mitigation methods. ✓ indicates that the mechanism is explicitly modeled; ● indicates partial or indirect consideration; ✗ indicates that the aspect is not explicitly addressed.

Method	Intervention Level	Primary Target	Head-aware	Head-specific Visual Cues	Visual Forgetting Modeling	Text-side Context Modeling	External Statistics/Priors
CODE [19]	Logits	Description contrast	✗	✗	✗	✗	✗
VCD [16]	Logits	Output distribution	✗	✗	✗	✗	✗
OPERA [18]	Decoding / Attention	Over-trusted attention	✗	✗	✗	✗	✗
DoLa [17]	Logits	Layer-wise contrast	✗	✗	✗	✗	✗
DropoutDecoding [20]	Logits	Uncertainty-aware decoding	✗	✗	✗	✗	✗
VAF [21]	Attention	Visual/system tokens	✗	✗	✗	✗	✗
SPARC [24]	Attention	Aggregated visual attention	✗	✗	●	✗	✗
VAR [22]	Attention	Visual attention sink	●	✗	●	✗	●
VHR [23]	Attention	Vision-sensitive heads	✓	✗	✗	✗	✗
AdaIAT [25]	Attention	Generated-text attention	✓	✗	✗	●	✓
Modular Attribution [43]	Attention	Hallucination-related heads	✓	✗	✗	●	✓
VerDA	**Attention**	**Head-wise visual cues**	✓	✓	✓	✓	✗

G Details on the GPT-4o Evaluation

G.1 Evaluation on CapMAS

To evaluate the performance of LVLMs on CapMAS, we adopt GPT-4o as the evaluator following the official evaluation protocol of CapMAS [41]. For each image, we first generate model responses using five semantically equivalent detailed-captioning prompts: "Describe the given image in a very detailed manner", "Provide a detailed description of the specified image", "Elaborate on the details of the image provided", "Offer an in-depth description of the given image", and "Thoroughly describe the features of the specified image". Following the CapMAS protocol, the five responses are merged into a single detailed caption using Meta-LLaMA-3-8B-Instruct. The prompt used for caption merging is shown in Table 16. The resulting caption is evaluated by GPT-4o, which makes

Table 16: Prompt template used for caption merging with Meta-LLaMA-3-8B-Instruct.

Role	Prompt
System	This is a hard problem. Carefully summarize in ONE detailed caption based on the following 5 captions by different people describing the same image. Be sure to describe everything, and avoid hallucination. Provide the detailed caption in the format {Detailed caption}.
User	Caption 1: {generated_captions[0]} Caption 2: {generated_captions[1]} Caption 3: {generated_captions[2]} Caption 4: {generated_captions[3]} Caption 5: {generated_captions[4]}

the evaluation more robust to prompt variation and reduces evaluation cost. We report three metrics provided by CapMAS: *Factuality*, *Coverage*, and *CLAIR*. For factuality, GPT-4o receives the image, the reference description, and the decomposed propositions, and determines whether each proposition is true or false. For coverage, GPT-4o answers multiple-choice questions using only the generated caption. For CLAIR [49], GPT-4o compares the candidate caption set with the reference caption set and outputs a score from 0 to 100 in JSON format. The prompt templates used for GPT-4o evaluation on CapMAS are shown in Table 17.

Table 17: Prompt templates used for GPT-4o evaluation on CapMAS.

Metric	Prompt Template
Factuality	**System:** Your role is to determine whether the given propositions are True or False based on the provided image and its description. The outputs should follow the following format: "1. True/False 2. True/False 3. True/False 4. True/False 5. True/False ...". The number of True/False answers must match the number of propositions. **User:** Description: {caption}. Propositions: {questions} **Image:** {image}
Coverage	**System:** Your role is to answer the given questions based on the provided caption. I want to measure the amount of information in the caption. Therefore, if the correct answer to the question cannot be determined from the caption, you should answer it with "I don't know". Do not use your own knowledge in your response. Do not use information that can be inferred from the question itself. Only use the information provided in the caption. Answer the question by directly selecting the letter of the corresponding option. Do not repeat the question. **User:** Caption: {caption}. Questions: {questions}
CLAIR	You are trying to tell if a candidate set of captions is describing the same image as a reference set of captions. Candidate set: {candidate captions}. Reference set: {reference captions} On a precise scale from 0 to 100, how likely is it that the candidate set is describing the same image as the reference set? JSON format, with a key "score", value between 0 and 100, and a key "reason" with a string value.

Table 18: The prompt used for GPT-4o evaluation on LLaVA Bench.

GPT-4o Prompt
You are required to score the performance of two AI assistants in describing a given image. You should pay extra attention to the hallucination, which refers to the part of descriptions that are inconsistent with the image content, such as claiming the existence of something not present in the image or describing incorrectly in terms of the counts, positions, or colors of objects in the image. Please rate the responses of the assistants on a scale of 1 to 10, where a higher score indicates better performance, according to the following criteria: 1: Accuracy: whether the response is accurate with respect to the image content. Responses with fewer hallucinations should be given higher scores. 2: Detailedness: whether the response is rich in necessary details. Note that hallucinated descriptions should not count as necessary details. Please output the scores for each criterion, containing only two values indicating the scores for Assistant 1 and 2, respectively. The two scores are separated by a space. Following the scores, please provide an explanation of your evaluation, avoiding any potential bias and ensuring that the order in which the responses were presented does not affect your judgment. [Assistant 1] {} [End of Assistant 1] [Assistant 2] {} [End of Assistant 2] Output format: Accuracy: Reason: Detailedness: Reason:

G.2 Evaluation on LLaVA-Bench

To evaluate the performance of LVLMs on LLaVA-Bench, we adopt GPT-4o as the evaluator. For each sample, we follow the prompt template shown in Table 18, as introduced in [51], and provide GPT-4o with the original image, the baseline LVLM output, and the corresponding output enhanced by VerDA. The evaluation focuses on assessing both the accuracy and the level of detail in the generated responses, with particular emphasis on the reduction of hallucinations achieved by VerDA compared to the baseline. To mitigate potential positional bias in the evaluator, we observe that GPT-based models tend to favor the first response when comparing multiple candidates. Therefore, we randomize the order of the baseline and VerDA outputs and repeat the evaluation four times. The final scores are obtained by averaging the results across all runs. Figures 13, 14, and 15 illustrate representative evaluation examples based on the three model architectures: LLaVA-1.5-7B, LLaVA-Next-8B, and InstructBLIP-13B, respectively.

H Compute Resources

All experiments are conducted on a server equipped with:

- CPU: Intel Xeon Silver 4316 @ 2.30GHz × 2
- GPU: NVIDIA RTX A6000 48GB × 8

I Broader Impacts

This work aims to mitigate hallucinations in large vision-language models (LVLMs). Specifically, we propose a training-free method that leverages attention head heterogeneity and modulates the attention to mitigate hallucinations, thereby improving consistency between generated content and visual inputs. Our approach has the potential to enhance the trustworthiness of multimodal systems in real-world applications, such as visual question answering, image captioning, and assistive technologies, by reducing the generation of misleading content. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

[Input Image]



[Question] This is the logo of LLaVA, Large Language and Vision Assistant, based on the LLaMA architecture. Please explain this logo in detail, and how do you think of its design.

[Model Outputs]

LLaVA-1.5-7B

The LLaVA logo features a red and orange fire-like creature with a pair of glasses on its face. The creature appears to be a mix of a horse and a dragon, with a flaming mane and legs. The glasses on the creature's face give it a quirky and unique appearance.\n\nThe design of the LLaVA logo is visually striking and memorable. The combination of the fire-like creature and the glasses creates a sense of whimsy and playfulness. The choice of colors, red and orange, adds to the vibrant and energetic nature of the logo. Overall, the LLaVA logo is a creative and eye-catching representation of the brand's vision and values.

LLaVA-1.5-7B w/VerDA

The LLaVA logo features a red and orange fire-like creature with a pair of glasses on its face. The creature appears to be a mix of a horse and a dragon, with a flame-like design on its legs. The glasses on the creature's face give it a quirky and playful appearance. The design of the logo is visually striking and memorable, making it easy to recognize and associate with the LLaMA architecture. The combination of the fire-like creature, the glasses, and the flame-like design on the legs creates a unique and engaging visual representation of the LLaMA architecture.

GPT-4o Evaluation

Accuracy: **6.0 10.0**
Reason: Assistant 1: The description is accurate overall, identifying the bear, cat, and rabbit sitting around a table and enjoying a meal together, with a plate of food present. However, it makes a slight generalization about the behavior of all animals eating, while the image clearly shows only the bear actively eating. Assistant 2: This description is fully accurate. It identifies the three animals and their actions: the bear holding a spoon, the cat holding a cup, and the rabbit drinking from a pink cup.
Detailedness: **7.0 9.0**
Reason: Assistant 1: While Assistant 1 gives an accurate high-level description of the scene, it lacks specific details about the objects (e.g., the spoon and the pink cup) and individual behaviors of the animals. Assistant 2: Assistant 2 provides more precise details, such as identifying specific objects (spoon and cup) and the animal behaviors associated with them, which enhances the richness of the description.

Figure 13: The performance of VerDA on mitigating hallucinations on LLaVA-1.5-7B.

[Input Image]



[Question] What's the personality of this character? Explain what elements or aspects of the character's design may have contributed to its popularity.

[Model Outputs]

LLaVA-Next-8B

The character in the image appears to be from the anime series \"JoJo's Bizarre Adventure.\" The character's design and personality are often associated with the Stand powers and the overall aesthetic of the series, which is known for its unique and flamboyant characters. The character's large, expressive eyes and the distinctive hairstyle with a parting in the middle are typical of anime and manga art styles. The character's outfit, which includes a suit and a bow tie. The Stand powers are a key aspect of the series' appeal. The Stand powers often reflect the personality and style of the character, and they are often visually striking and memorable. The series has a dedicated fan base and has been widely popular both in Japan and internationally.

LLaVA-Next-8B w/VerDA

The character in the image appears to be a young man with a somewhat mischievous and curious expression. His large, expressive eyes and the glasses he's wearing give him a somewhat intellectual or studious look. The red bow tie adds a touch of formality to his outfit. The character's design elements that may have contributed to his popularity include:\n\n1. His large, expressive eyes which are a common trope in anime and manga, often used to convey emotion and personality.\n2. His glasses, which add a touch of sophistication and intellectualism.\n3. His confident and curious expression, which makes him appear to be a well-rounded character. 4. His outfit, which is a mix of formal and casual, suggesting a character who is both stylish and approachable. Overall, the character's design elements work together to create a character that is both visually appealing and intriguing, which are key factors in making a character popular.

GPT-4o Evaluation

Accuracy: **3.0 9.0**
Reason: Assistant 1 includes significant inaccuracies by associating the character with \"JoJo's Bizarre Adventure\" and mentioning \"Stand powers,\" which are irrelevant to the character in the image. Assistant 2 provides an accurate description of the character's physical traits, such as glasses, red bow tie, hairstyle, and expression, without introducing unrelated elements.
Detailedness: **5.0 8.0**
Reason: Assistant 1: Assistant 1 discusses the general themes of \"JoJo's Bizarre Adventure\" and its character design philosophy but is unrelated to the image, reducing its relevance. Assistant 2 provides a more focused, detailed breakdown of the character's features while staying consistent with the image.

Figure 14: The performance of VerDA on mitigating hallucinations on LLaVA-Next-8B.

[Input Image]



[Question] Describe this painting in detail.

[Model Outputs]

Instrutblip-13B

The painting portrays a close-up of a woman's face, which is commonly known as the Mona Lisa. Her long hair is tied back, and she has a calm expression. The painting is a portrait of the woman, with her eyes and lips clearly visible. The woman is wearing a black dress, and her hair is long and dark. The painting is a masterpiece, and its beauty is evident in the attention to detail and the skillful execution of the portrait. <s>

Instrutblip-13B w/VerDA

The painting portrays a close-up of a woman's face, which is believed to be the famous Mona Lisa. Her long hair is flowing down her shoulders, and her eyes are focused on the viewer. The painting is a portrait of the woman, and her facial expression is calm and serene. The painting is a masterpiece, and its beauty and intricate details make it a popular and iconic work of art. <s>

GPT-4o Evaluation

Accuracy: **5.0 8.0**
Reason: Both assistants recognize the painting as the Mona Lisa, confirming the depicted content corresponds to the image. However, Assistant 1 inaccurately states that the woman's hair is tied back, which contradicts the visible flowing hair in the painting and introduces hallucination. Assistant 2 correctly describes the hair as flowing down the shoulders, aligning with the image.
Detailedness: **4.0 7.0**
Reason: Assistant 1: Assistant 1 provides foundational observations such as the calm expression, visible facial features, and mentions the black dress. However, it misses out on describing elements like the flowing background scenery. Assistant 2 captures slightly more detail of the scene, emphasizing the serene expression and intricate details, though both assistants could have elaborated on the broader landscape behind the subject for richer detail.

Figure 15: The performance of VerDA on mitigating hallucinations on InstructBLIP-13B.

J Ethical Consideration

VerDA aims to reduce hallucinations in LVLMs by reinforcing head-adaptive visual cues and encouraging more reliable visual grounding. This can improve the factuality and trustworthiness of multimodal outputs, aligning with ethical principles such as safety, reliability, and transparency. Since VerDA is training-free, it does not require additional data collection or model fine-tuning, thereby avoiding extra privacy risks from new training data. Nevertheless, as with existing hallucination mitigation methods, VerDA does not guarantee hallucination-free generation. Although our experiments provide extensive validation across multiple LVLMs and benchmarks, VerDA should still be applied with caution in real-world scenarios. In safety-critical applications, model outputs should be carefully verified and, where appropriate, reviewed by humans or domain experts.

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